Feiyang Huang

PhD Candidate, Computational Biology & Medicine

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fyng · feiyang-huang · English, Mandarin

Education

PhD **Weill Cornell Graduate School of Medical Sciences**, Tri-Institutional PhD Program in Computational Biology & Medicine — New York, NY — Aug 2023 – present
- Advisor: Wesley Tansey, Memorial Sloan Kettering Cancer Center

BS **Johns Hopkins University**, Computer Science and Biomedical Engineering — Baltimore, MD — Sept 2020 – May 2023
- GPA 3.96
- Activities: President, Singapore Students' Association (2021–2023); Theta Tau Professional Engineering Fraternity (2020–2022)

Experience

Memorial Sloan Kettering Cancer Center, Tansey Lab, PhD Researcher — New York, NY — June 2024 – present
- Developing OncoState, a self-supervised transformer learning patient states from multimodal clinical trajectories of 99,000+ MSK-IMPACT patients; outperforms the IMDC risk model in advanced clear cell RCC.
- Co-led a population-scale study of clinical and genetic determinants of systemic-therapy toxicity (co-first author, medRxiv 2026): developed the random forest toxicity model and the analysis and narrative for Figure 3.
- Contributed to PPC, a pan-cancer atlas of ex vivo drug screens for functional precision oncology.

Memorial Sloan Kettering Cancer Center, PhD Rotations — New York, NY — Aug 2023 – May 2024
- **Tansey Lab**: Bayesian active learning for nominating combination therapies from ex vivo drug screens
- **Chodera Lab**: Equivariant graph neural networks for small-molecule property prediction (mtenn)
- **CMO Innovation Lab**: Computational pipeline for bisulfite- and enzyme-based DNA methylation sequencing of tumors
- **Morris Lab**: BERT representation learning from tumor mutation profiles

Johns Hopkins University, Fan Lab, Undergraduate Researcher — Baltimore, MD — Jan 2021 – May 2023
- Built and benchmarked a reference-free LDA pipeline to deconvolve spatial transcriptomics data (STdeconvolve, *Nature Communications* 2022).
- Developed a torque-based method and simulation framework to quantify subcellular mRNA polarization.

Hologic, Software Engineering Intern — May 2022 – Aug 2022
- Built an internal work-history search tool and automated release-note generation (3 h to 20 min).

Johns Hopkins University, Biomedical Design, Team Lead, iMEDS — Baltimore, MD — Aug 2022 – May 2023
- Led a team of 7 building an algorithm to continuously monitor pediatric ICU sedation; interviewed 20+ clinicians.

Johns Hopkins University, Institute for NanoBioTechnology, Research Intern — Baltimore, MD — May 2021 – Aug 2021
- Optimized a point-of-care PCR assay (15% lower detection limit) and its embedded image-analysis and motor-control code.

DioTeX Diagnostics, Co-Founder — Baltimore, MD — Nov 2020 – Aug 2023
- Led assay development for a lateral-flow test for internal hemorrhage; raised over $60,000 in non-dilutive funding.

National University of Singapore, Yu Lab, Research Intern — Singapore — Nov 2019 – Dec 2020
- Trained a CNN for endoscopic image classification (91% accuracy); co-author in *Gastrointestinal Endoscopy*.

Singapore Armed Forces, 3rd Sergeant — Singapore — Jan 2018 – Oct 2019
- Led a detachment of 3 soldiers and co-supervised daily training for 15.

Publications

Population-Scale Precision Safety in Oncology Reveals Clinical and Genetic Determinants of Systemic Therapy Toxicity — Sept 2026
Ziad Bakouny*, X Alex Guo*, **Feiyang Huang***, Saksham Mohan*, Rohan Walser*, et al., Wesley Tansey, Jian Carrot-Zhang, Ed Reznik
www.medrxiv.org/content/10.64898/2026.09.16.26363259 (medRxiv preprint; *equal contribution)

A Pan-Cancer Ex Vivo Drug Screen Atlas for Functional Precision Oncology — Feb 2026
Karl Pichotta, Jessica B White, Jeffrey F Quinn, Anneliese Markus, Christopher Tosh, Antoine De Mathelin, Erin Coyne, **Feiyang Huang**, Wesley Tansey
pmc.ncbi.nlm.nih.gov/articles/PMC12934811 (bioRxiv preprint)

Reference-free cell type deconvolution of multi-cellular pixel-resolution spatially resolved transcriptomics data — Apr 2022
Brendan F Miller, **Feiyang Huang**, Lyla Atta, Arpan Sahoo, Jean Fan
10.1038/s41467-022-30033-z (Nature Communications 13, 2339)

Artificial intelligence-enhanced white-light colonoscopy with attention guidance predicts colorectal cancer invasion depth — Sept 2021
Xiaobei Luo, Jiahao Wang, Zelong Han, Yang Yu, Zhenyu Chen, **Feiyang Huang**, et al., Hanry Yu
10.1016/j.gie.2021.03.936 (Gastrointestinal Endoscopy 94(3), 627–638)

Roles and regulation of long noncoding RNAs in hepatocellular carcinoma — Oct 2019
Lee Jin Lim, Samuel YS Wong, **Feiyang Huang**, et al., Caroline G Lee
10.1158/0008-5472.CAN-19-0255 (Cancer Research 79(20), 5131–5139)

Conference Abstracts

- Bakouny Z, Guo XA, Walser R, **Huang F**, et al. Pan-cancer discovery of clinical and genomic determinants of immune-related adverse events. *Journal of Clinical Oncology* 44(16_suppl):11132 (ASCO 2026).
- Wu S, Pejic B, Roy S, **Huang F**, et al. Development of a computational model of SBS scores in critically ill children. *Pediatric Critical Care Medicine* 25(11S):e87 (2024).

Patents

- Parent C, Simon E, **Huang F**, Ruci A, Zhang E. Methods and devices for quantitatively estimating thrombomodulin. US Patent App. 19/240,058 (2025).
- Eng R, Parent C, Hayat A, Ruci A, Krishnan A, **Huang F**, Simon E, Zhang E. Methods and devices for quantitatively estimating syndecan-1. US Patent App. 17/715,294 (2022).

Presentations

- 45th Vincent du Vigneaud Research Symposium, Weill Cornell Medicine, New York, NY (2026). **Huang F**, Tansey W. OncoState: learning patient states from clinical trajectories in cancer. Poster #66.
- Military Health System Research Symposium, Kissimmee, FL (2022). Towards point-of-care internal hemorrhage detection through a syndecan-1 lateral flow assay. Oral presentation.
- Johns Hopkins BME Design Day, Baltimore, MD (2022). DioTeX: Hemorrhage Diagnostics. Poster.

Honors and Awards

NIH DEBUT Challenge, Healthcare Technologies for Low-Resource Settings Prize — 2023
With the DioTeX team; $15,000.

VentureWell e-Fest Prize — 2023
With the DioTeX team; $10,000.

Teaching

- Course Assistant, EN.601.230 Mathematical Foundations for Computer Science, Johns Hopkins University (Fall 2022).